

health impact functions, move the first category on which you wish to sort to the first position after the **Pooling Method** column. In this example, you would drag the **Author** column ahead of **Endpoint** column. As you can see, after this change the pooling options are very different.

Sorted by Author:

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: PoolingWindow2 Pooling Levels: 1 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

PoolingWindow2 x +

Studies, By Endpoint	Pooling Method	Nickname	Author	Endpoint	Qualifier	Study Location	Start Age	End Age	Age Range
Asthma Exacerbation	None	Asthma Exacerbation	Ostro et al.; Mar et al.	Asthma Exacerbation, Wheeze Asthma Ex...			6	18	6-18
Ostro et al.	Fixed Effects	Ostro et al.	Ostro et al.	Asthma Exacerbation, Wheeze Asthma Ex...			6	18	6-18
Ostro			Ostro et al.	Asthma Exacerbation, Wheeze	Ostro stud...	Los Ang...	6	18	6-18
Ostro			Ostro et al.	Asthma Exacerbation, Shortness of Breath	Ostro stud...	Los Ang...	6	18	6-18
Mar et al.	None	Mar et al.	Mar et al.	Asthma Exacerbation, Shortness of Breat...			6	18	6-18
Mar			Mar et al.	Asthma Exacerbation, Shortness of Breath	Uses incid...	Spokan...	6	18	6-18
Mar			Mar et al.	Asthma Exacerbation, Cough	Uses incid...	Spokan...	6	18	6-18

Preview Advanced Cancel Next

Sorted by Endpoint:

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: PoolingWindow1 Pooling Levels: 1 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

PoolingWindow1 x +

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study Location	Start Age	End Age	Age Range
Asthma Exacerbation	None	Asthma Exacerbation	Asthma Exacerbation, Shortness of Breat...	Mar et al.; Ostro et al.			6	18	6-18
Asthma Exacerbation, Shortness...	Fixed Effects	Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	Mar et al.; Ostro et al.			6	18	6-18
Mar			Asthma Exacerbation, Shortness of Breath	Mar et al.	Uses incid...	Spokan...	6	18	6-18
Ostro			Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18
Asthma Exacerbation, Wheeze		Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.			6	18	6-18
Ostro			Asthma Exacerbation, Wheeze	Ostro et al.	Ostro stud...	Los Ang...	6	18	6-18
Asthma Exacerbation, Cough		Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Mar et al.			6	18	6-18
Mar			Asthma Exacerbation, Cough	Mar et al.	Uses incid...	Spokan...	6	18	6-18

Preview Advanced Cancel Next

If you use two different pooling windows, each sorted as shown above, you can create results pooled by **Author**, and results pooled by **Endpoint**.

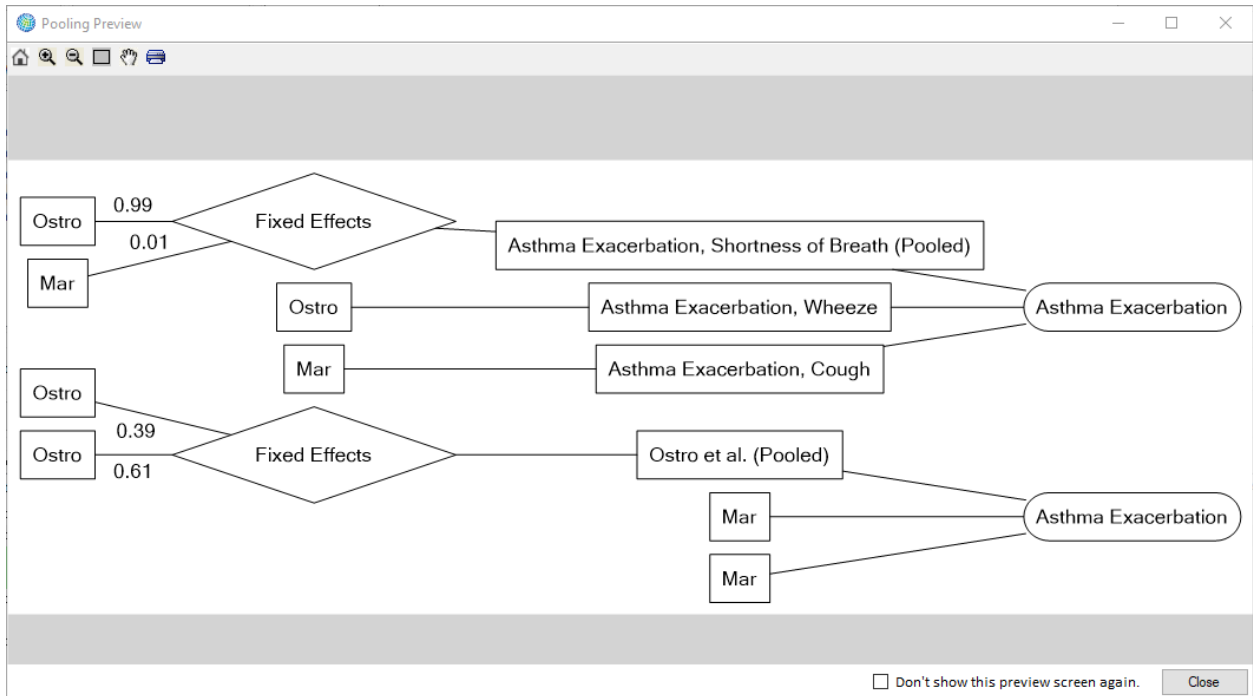
7.2.3 Valuing Pooled Incidence Results

After you have specified your incidence pooling options, click on the **Next** button and BenMAP-CE will show a **Pooling Preview**. Review and **Close** the **Pooling Preview** in order to select valuations and valuation pooling options from the **Select Valuation Methods, Pooling and Aggregation** window.

BenMAP-CE Decision Point

When might you want to pool your valuation estimates?

- You have multiple valuation functions (likely derived from different studies) for the same health endpoint that you would like to average.
- For a given endpoint, there are age-specific valuation functions for non-overlapping age groups that you would like to sum to obtain a total valuation estimate.



The **Select Valuation Methods, Pooling and Aggregation** window should look quite similar to the **Select and Pool Incidence Estimates** window, with tree views on the top half representing the valuation methods available, and various pooling windows on the lower half representing the selected valuations and pooling options.

BenMAP-CE Use Example

Example scenarios for pooling valuation estimates:

- You have two valuation functions derived from different studies for *Emergency Room Visits, Asthma* that you would like to average using equal or user-specified weights.
- You have multiple age-specific valuation functions for *Acute Myocardial Infarction* (for age groups 0-24, 25-44, 45-54, 55-64, and 65-99) that you would like to combine into a single valuation estimate for ages 0-99.

For each pooling window you created on the **Select and Pool Incidence Estimates** window, there is a corresponding pooling window in the **Select Valuation Methods, Pooling, and Aggregation** window. You will notice that the number of incidence estimates available in the valuation pooling window will reflect any pooling choices you made in the **Incidence Pooling and Aggregation** window. For example, if in the **Select and Pool**

Incidence Estimates window you pooled 5 incidence estimates into a single incidence estimate, you will see a single incidence estimate in the **Valuation Methods Pooling and Aggregation** window.

The columns present in the **Select Valuation Methods, Pooling, and Aggregation** window are determined by the incidence results left after all incidence pooling has occurred. There will be exactly enough columns in each pooling window to represent the “least” pooled incidence result. That is, the columns will be in the same order they were in the **Select and Pool Incidence Estimates** window, but the only columns

present will be those up to the level of the pooled incidence result with the most columns left over after all pooling has occurred. Here is an example:

1. Select Valuation Method

Incidence results for individual studies and pooled groups are listed in the bottom window. To assign a valuation function to a given set of incidence results, click and drag the valuation function(s) from the top window to the bottom window and release in the highlighted orange "drop zone" for the desired study or pooled group.

EndPoint	Endpoint Group	Start A...	End ...	Function	Dataset N...	Qualifier	Refere...	A	NameA	DistA	P1A	P2A	B	NameB	C	NameC	D	NameD
Asthma E...	Asthma Exacerb...	18	99	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	18	99	A*B*All...	Additional ...	WTP: 1 s...		104.068...	mean o...	LogNo...	4.64047...	0.09568...	0.97381...	adjustm...	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*B*All...	Additional ...	WTP: 1 s...		220.910...	mean o...	LogNo...	5.39347...	0.09253...	0.97381...	adjustm...	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*B*All...	Additional ...	WTP: 2 x...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	2	adjustm...	0	0	0	
Asthma E...	Asthma Exacerb...	18	99	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	18	99	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*AllGo...	Additional ...	WTP: ba...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	0	0	0	0	0	
Asthma E...	Asthma Exacerb...	0	17	A*B*All...	Additional ...	WTP: 2 x...		58.9006...	WTP in ...	Uniform	21.4167...	97.5632...	2	adjustm...	0	0	0	
Asthma E...	Asthma Exacerb...	18	99	A*B*All...	Additional ...	WTP: 1 s...		104.068...	mean o...	LogNo...	4.64047...	0.09568...	0.97381...	adjustm...	0	0	0	

2. Organize and Show Selections

Variable Dataset: Delete Selected Function(s)

PoolingWindow1 | PoolingWindow2

Studies, By Endpoint	Nickname	No. of Studies	Pooling Method	Author	Qualifier	End Po...	Author	Qualifier	Colum...
Asthma Exacerbation	Asthma Exacerbation	4		Mar et al.; Ostro et al.		Asthma ...	Mar et ...		
Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	2		Mar et al.; Ostro et al.		Asthma ...	Mar et ...		
Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	1		Ostro et al.		Asthma ...	Ostro et...		
Ostro		0		Ostro et al.	Ostro study po...	Asthma ...	Ostro et...	Ostro st...	
Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	1		Mar et al.		Asthma ...	Mar et al.		
Mar		0		Mar et al.	Uses incidenc...	Asthma ...	Mar et al.	Uses in...	

Resolve Pooling Conflict

Advanced Cancel Save As (*.apvx) Run As (*.apvx)

There are several steps to take in the **Select Valuation Methods, Pooling, and Aggregation** window:

Step 1. Select your valuation methods

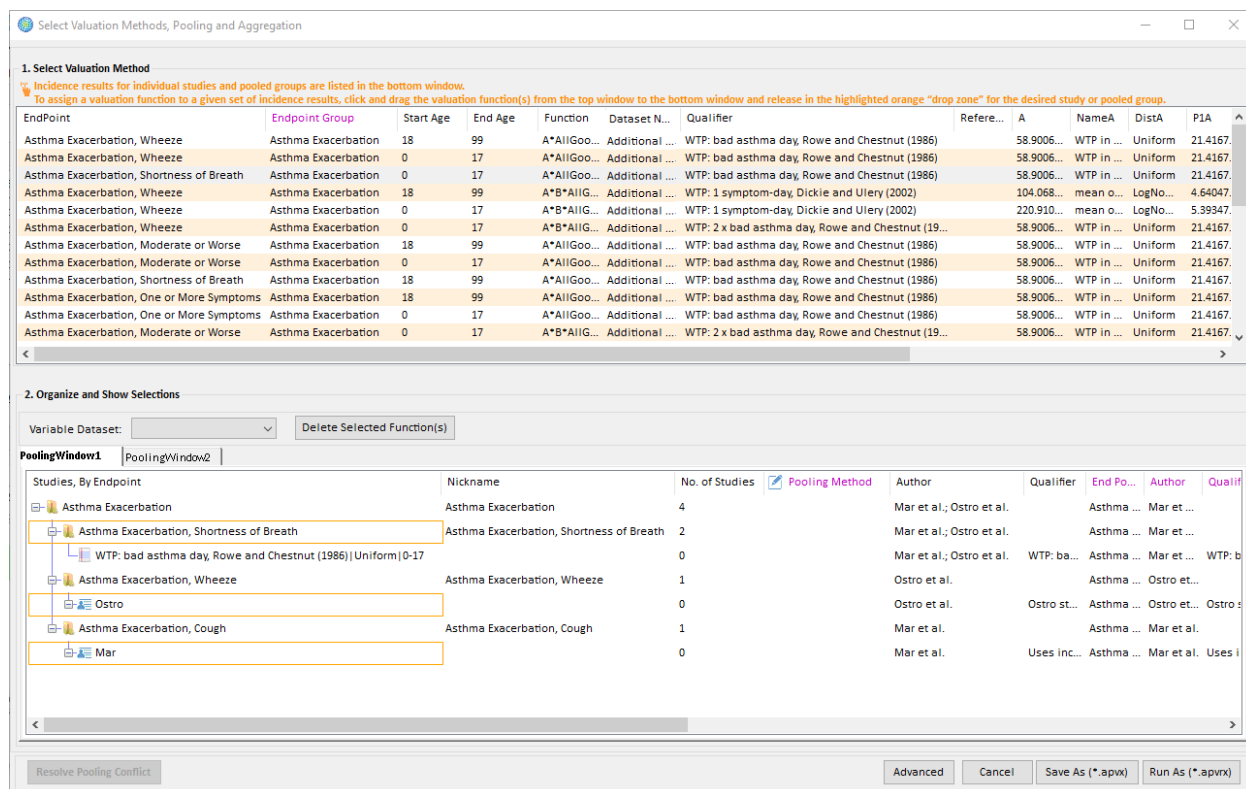
Valuation Methods are specific to **Endpoint Groups**, and sometimes to **Endpoints** as well. The only **Valuation Methods** which appear on the top half of the window are those which have the same **Endpoint Group** values as the pooled incidence results which are available to be valued. To select a **Valuation Method**, select it from the table and drag-and-drop it onto the appropriate incidence result in the pooling window below. Note that BenMAP-CE will only allow you to drop **Valuation Methods** onto incidence results which have the same **Endpoint Group** value. For example, BenMAP-CE will not allow you to drop a 'Mortality' valuation on a 'Hospital Admissions' incidence result. Note also that you can only drag-and-drop individual **Valuation Methods**, not entire groups of them. For explanations of the various valuation methods, see Appendix G.

BenMAP-CE Guidance and Best Practices

When selecting Valuation Methods for your analysis, it is important to match the valuation function to the health incidence estimate as closely as possible. For example, if the health endpoint is *Asthma Symptoms*, *Albuterol use*, then the valuation function should correspond specifically to albuterol use rather than other asthma symptoms such as cough or wheeze.

For long-term health impacts, the valuation function may also account for your preferred time span of analysis and discount rate.

If you have added any of your own valuation methods, as described in the Valuation Data section of Chapter 4: Loading Data, you can drag-and-drop them in the same way as the *EPA Standard Valuation Functions* shown in these examples.



When BenMAP-CE runs the APV Configuration, it will generate a valuation result for each **Valuation Method** you select by running the method's **Function** on the selected incidence results. You do not need to select a **Valuation Method** for every incidence result—incidence results without any **Valuation Methods** will simply be ignored when valuation results are generated, aggregated, and pooled.

Because valuation functions include an uncertainty distribution around them, generating valuation results is fairly complicated. The procedure depends on whether the incidence results being used were generated in Point Mode or with the default Monte Carlo method. See Chapter 6: Incidence Estimation (Section 6.2.3) for details on these advanced configuration settings.

In Point Mode, BenMAP-CE simply runs the valuation functions once using the point estimate of the incidence result and the mean of the valuation function as inputs.

With the Monte Carlo feature, on the other hand, BenMAP-CE generates one hundred percentile points (from the 0.5th percentile to the 99.5th percentile) to represent the distribution of the inputs to the valuation function. To get the value of the health incidence, BenMAP-CE multiplies each combination of values from the incidence result with each of the hundred valuation points, and puts the results into a holding container. (For example, if the incidence result has 10 percentile points and there are 100